

A Blood bank is a place, for carrying out all or any of the operations for collection, aphaeresis', storage, processing and distribution of blood drawn from donors and/or for preparation, storage and distribution of blood components. By an organization. One of the earliest blood bank was established by Frederic Durán-Jordà during the Spanish Civil War in 1936. A blood collection program was initiated in the US in 1940.

Someone needs blood every two seconds. About 1 in 7 people entering a hospital need blood. Healthy adults who are at least 17 years old, and at least 110 pounds may donate about a pint of blood—the most common form of donation—every 56 days, or every two months. Females receive 53 percent of blood transfusions; males receive 47 percent.

Dr. Karl Landsteiner first identified the major human blood groups – A, B, AB and O – in 1901. One unit of blood can be separated into several components: red blood cells, plasma, platelets and cryoprecipitate. Red blood cells live about 120 days in the circulatory system. Healthy bone marrow makes a constant supply of red cells, plasma and platelets. Donated red blood cells can be stored for 42 days.

Donated platelets can be stored for five days. Frozen plasma can be stored for one year. According to the Central Drug Standard Control Organisation (CDSCO), India has 2,760 licensed blood banks. Maharashtra (308) had the highest number of blood banks followed by Tamil Nadu (265), Uttar Pradesh (248), Karnataka (185), Kerala (166), Telangana (153), Gujarat (134), Madhya Pradesh (133), Andhra Pradesh (125), West Bengal (115) and Rajasthan (102).



1.3 Musculo skeletal system

It consists of

- Bones
- Muscles
- Cartilage
- Tendons
- Ligaments
- Joints
- Facia

Bones

Bone tissue is a hard tissue, a type of dense connective tissue. Bone tissue is made up of different type of bone cells.

Bone cells

Osteocytes are the building blocks of the bone and form the inner matrix of bone tissue that gives strength to bones.

Osteoclasts maintain the integrity and strength of the bones.

Osteoblasts build new bones by producing collagen and minerals so that the osteocyte can remain in good functioning condition.

Stem cells form in the inner surface of the bone will later transform into osteoblasts.

Lining cells protect the bones and they also release calcium into the blood when the blood calcium levels are low.

Types of the bones

1. **Long bones:** Present in the arms and legs. They act as levers to move parts
2. **Flat bones:** These includes the ribs, scapulae, sternum, and bones of the cranium
3. **Irregular bones:** These bones are seen in face and spine bones.
4. **Short bones:** These bones are seen in wrist and ankle bones.

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